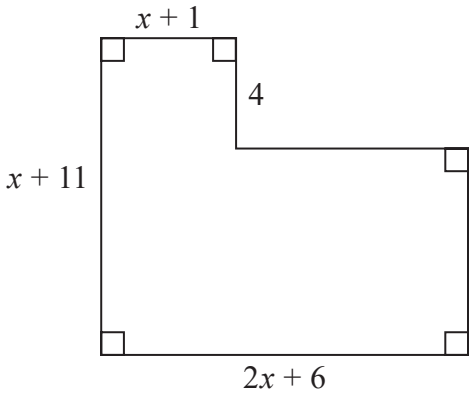
**14** Here is a shape with all its measurements in centimetres.




The area of the shape is $A$ cm$^2$

Show that $A = 2x^2 + 24x + 46$

Outer rectangle: $(2x + 6)(x + 11)$

Missing top-right rectangle: $4(x + 5)$

$A = (2x + 6)(x + 11) - 4(x + 5)$

$= 2x^2 + 28x + 66 - 4x - 20$

$= 2x^2 + 24x + 46$, as required.

**1 mark**
one correct rectangle area

**1 mark**
complete subtraction

**1 mark**
expand to stated result

**(Total for Question 14 is 3 marks)**

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA